# BUSFIN 8200: Asset Pricing Foundations
# Problem Set 2

October 20, 2026

## Question 1

## Question 2

### 2a)

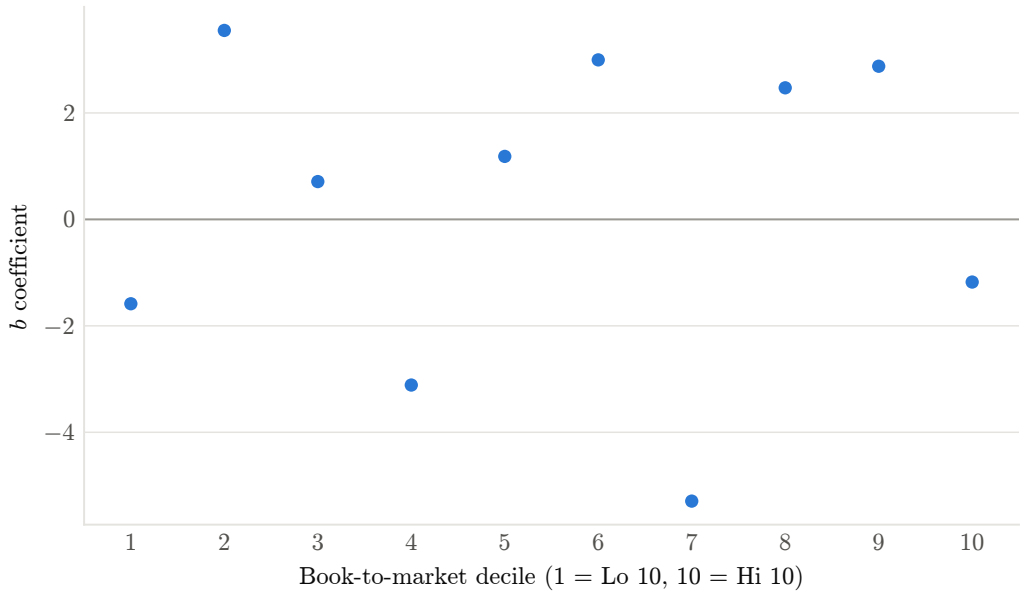


Figure 1: Coefficients $b$ of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 10 value-weighted book-to-market decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

### 2b)

The sample (Pearson) correlation between the monthly SDF series $M^*_{\mathrm{BM},t}$ estimated in 2a) and $M^*_{\mathrm{MOM},t}$ estimated in 2b), over July 1927 to August 2026 ($T = 1190$ months), is

$$\widehat{\mathrm{Corr}}\left(M^*_{\mathrm{BM},t},\, M^*_{\mathrm{MOM},t}\right) = 0.355.$$

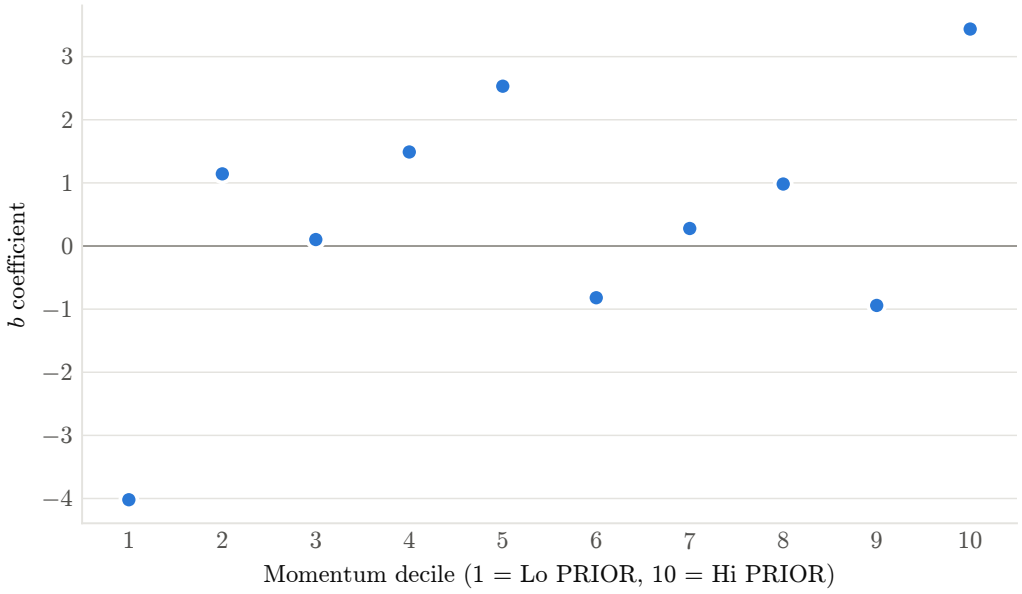


Figure 2: Coefficients $b$ of the SDF $M_t^* = a - b'xR_t$ that prices the monthly excess returns of the 10 value-weighted momentum (prior 12–2 month return) decile portfolios (July 1927 to August 2026), under the normalization $\widehat{\mathbb{E}}[M^*] = 1$. Excess returns are in decimal units.

## Question 3

## Question 4